

TERRA NOVA NATL PARK, NL, Canada

WMO: 715890

Lat: 48.560N

Lon: 53.970W

Elev: 350

StdP: 14.51

Time Zone: -3.50 (NAF)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.7	5.4	-6.7	3.9	3.7	-2.5	4.9	7.6	25.5	25.8	23.3	26.1	8.8	250	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.3	81.4	66.1	78.1	65.0	74.8	63.8	69.6	76.4	67.7	73.9	65.8	71.7	10.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	101.7	72.4	65.5	95.4	70.9	63.5	89.0	68.8	33.9	76.2	32.3	74.0	30.8	71.9	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.1	16.1	DB	-3.7	87.9	4.2	2.3	-6.7	89.6	-9.2	90.9	-11.6	92.2	-14.7	93.9	(4)
(5)			WB	-4.0	73.0	4.0	1.9	-6.9	74.4	-9.2	75.6	-11.5	76.6	-14.4	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	22.3	22.3	27.3	35.9	44.4	53.2	62.8	62.8	55.6	45.5	36.9	27.7		
	DBStd	15.99	8.28	8.30	7.06	6.25	6.84	7.79	6.67	5.69	6.98	6.86	7.11	8.10		
	HDD50	4307	857	776	704	423	203	52	3	1	24	177	397	690		
	HDD65	8680	1322	1196	1169	872	639	360	119	108	292	605	843	1155		
	CDD50	1210	0	0	0	1	29	149	401	399	191	36	4	0		
	CDD65	109	0	0	0	0	0	7	52	41	9	0	0	0		
	CDH74	857	0	0	0	0	10	106	425	251	65	0	0	0		
	CDH80	152	0	0	0	0	2	17	86	33	14	0	0	0		
	WSAvg	7.8	8.9	8.9	8.4	7.8	7.1	7.2	7.2	6.9	7.5	7.6	8.0	8.5		
	PrecAvg	49.6	4.3	4.3	4.2	3.5	3.6	3.5	3.8	4.0	4.4	4.9	4.7	4.9		
(15) Precipitation	PrecMax	58.7	8.5	5.7	8.5	6.5	6.8	6.5	7.2	6.3	8.0	8.7	9.8	9.9		
	PrecMin	38.2	2.0	1.4	2.2	1.7	1.4	1.3	1.8	1.7	1.7	2.3	1.4	1.2		
	PrecStd	4.8	1.5	0.9	1.5	1.3	1.3	1.4	1.3	1.1	1.7	1.8	2.0	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	46.6	49.0	60.2	73.5	82.3	86.2	84.1	81.2	68.0	59.2	52.6		
		MCWB	49.8	45.0	43.1	48.7	56.5	65.3	68.1	66.8	67.8	61.1	56.1	52.2		
	2%	DB	42.2	41.6	43.4	54.7	66.8	77.0	82.4	79.8	74.5	63.8	55.9	47.8		
		MCWB	41.4	39.8	39.4	47.9	54.2	62.5	66.8	65.6	64.4	59.7	54.2	47.0		
	5%	DB	37.3	37.1	39.9	50.7	61.9	72.2	79.0	76.6	70.2	60.2	52.7	42.9		
		MCWB	36.5	35.2	36.0	44.7	52.0	60.1	65.7	65.0	62.0	56.2	51.2	41.9		
	10%	DB	33.6	33.6	36.8	46.4	57.0	67.2	75.2	73.5	66.9	56.8	48.5	38.7		
		MCWB	32.6	32.1	33.8	41.7	49.8	57.8	64.1	63.8	60.7	53.2	46.9	37.8		
	0.4%	WB	49.8	45.6	45.3	53.3	61.0	69.6	72.4	71.7	70.0	64.3	57.3	52.4		
		MCDB	50.4	46.2	47.2	55.6	68.7	78.7	81.1	78.2	76.7	65.9	58.3	52.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.7	40.0	40.3	49.1	57.0	65.4	69.9	69.3	66.7	60.7	54.6	47.2		
		MCDB	42.1	41.2	42.0	52.8	62.7	72.5	77.6	74.4	71.6	62.5	55.7	47.7		
	5%	WB	36.5	35.6	36.8	45.5	53.4	61.8	67.9	67.3	64.4	57.0	51.2	42.2		
		MCDB	37.1	36.9	39.1	49.4	60.0	69.0	75.4	72.8	68.7	59.5	52.6	42.8		
	10%	WB	32.9	32.3	34.2	42.0	50.1	58.6	65.8	65.4	61.8	53.5	47.1	38.0		
		MCDB	33.4	33.2	36.0	45.5	56.2	66.0	72.1	71.3	65.8	56.2	48.3	38.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.1	14.5	15.5	17.9	19.0	19.2	17.3	17.3	13.9	12.2	11.5		
		MCDBR	17.2	18.3	17.8	22.4	27.4	27.7	25.1	22.5	21.7	18.2	17.0	16.5		
		MCWBR	17.4	17.5	14.0	15.0	16.4	14.9	12.1	11.5	13.1	13.7	16.3	16.5		
	5% WB	MCDBR	17.3	17.7	16.2	20.1	24.2	24.1	21.0	18.9	19.2	16.7	16.9	16.5		
		MCWBR	17.7	17.7	14.4	15.5	16.2	14.6	12.4	12.1	13.4	14.5	16.7	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.303	0.336	0.359	0.372	0.398	0.416	0.396	0.361	0.339	0.314	0.304		
	taud		2.273	2.354	2.312	2.362	2.394	2.344	2.343	2.396	2.480	2.502	2.490	2.330		
	Ebn at Noon		235	271	278	282	282	274	267	267	267	255	238	221		
	Edh at Noon		24	27	34	36	36	38	38	34	29	24	19	20		
(44) All-Sky Solar Radiation	RadAvg		308	560	916	1275	1502	1610	1649	1377	1065	640	358	233		
	RadStd		28	54	83	134	118	154	145	98	72	67	43	23		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page